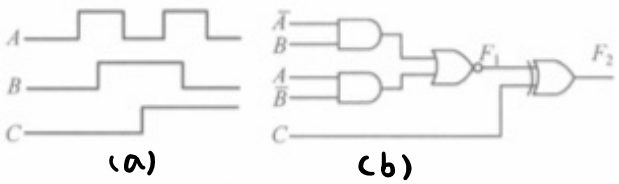
2.1

2.1　如果输入信号 A，B，C 的波形图如习题图 2.1(a)所示，试画出电路输出 F_1，F_2 的波形图。

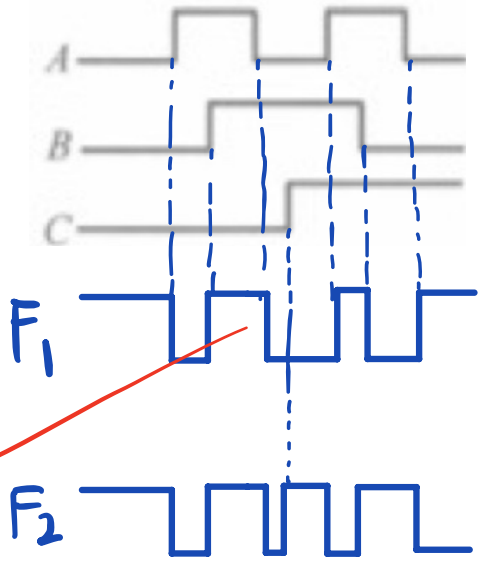


习题图 2.1

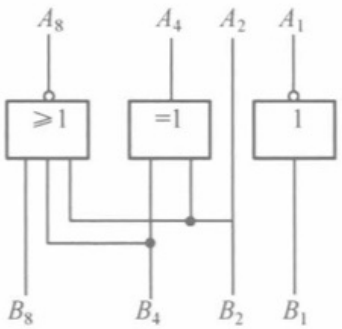
$$F_1 = \overline{\bar{A}B + A\bar{B}} = \overline{A \oplus B}$$

$$F_2 = F_1 \oplus C$$

$$= (\bar{A}B + A\bar{B})\bar{C} + \overline{(\bar{A}B + A\bar{B})}C$$



2.2



习题图 2.2

$$A_1 = \overline{B_1}$$

$$A_2 = B_2$$

$$A_4 = B_2 \oplus B_4$$

$$A_8 = \overline{B_2 + B_4 + B_8}$$

真值表：

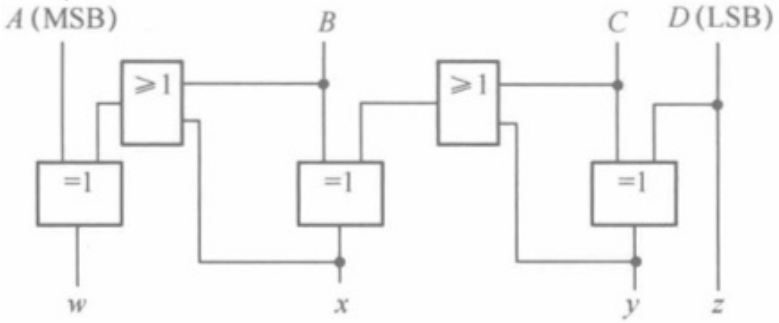
$B_8\ B_4\ B_2\ B_1$	$A_8\ A_4\ A_2\ A_1$
0 0 0 0	1 0 0 1
0 0 0 1	1 0 0 0
0 0 1 0	0 1 1 1
0 0 1 1	0 1 1 0
0 1 0 0	0 1 0 1
0 1 0 1	0 1 0 0
0 1 1 0	0 0 1 1
0 1 1 1	0 0 1 0
1 0 0 0	0 0 0 1
1 0 0 1	0 0 0 0
1 0 1 0	0 1 1 1
1 0 1 1	0 1 1 0
1 1 0 0	0 1 0 1
1 1 0 1	0 1 0 0
1 1 1 0	0 0 1 1
1 1 1 1	0 0 1 0

由真值表可知：

这是一个 BCD 码对 9 变补器

BCD码为0～9的十进制

2.3



习题图 2.3

真值表：

A B C D	w x y z
0 0 0 0	0 0 0 0
0 0 0 1	1 1 1 1
0 0 1 0	1 1 1 0
0 0 1 1	1 1 0 1
0 1 0 0	1 1 0 0
0 1 0 1	1 0 1 1
0 1 1 0	1 0 1 0
0 1 1 1	1 0 0 1
1 0 0 0	1 0 0 0
1 0 0 1	0 1 1 1
1 0 1 0	0 1 1 0
1 0 1 1	0 1 0 1
1 1 0 0	0 1 0 0
1 1 0 1	0 0 1 1
1 1 1 0	0 0 1 0
1 1 1 1	0 0 0 1

$Z = D$

$y = C \oplus D$

$x = B \oplus (C + y) = B \oplus (C + C \oplus D)$

$= B \oplus (C + D)$

$w = A \oplus (B + x) = A \oplus [B + (B \oplus (C + D))]$

$= A \oplus (B + C + D)$

由真值表可知：这是一个四位(16)二进制变补器

2.4 分别用与非门、或非门和与或非门设计如下电路：

(1) 3 变量的多数表决电路，以判断多数赞同；

输入：A, B, C　　输出：F

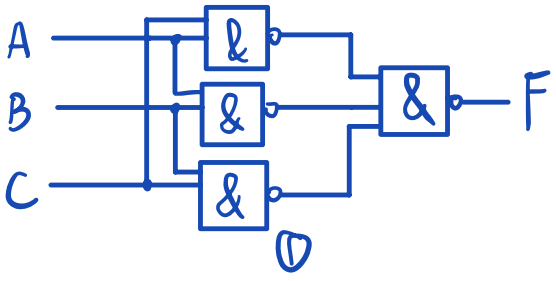
效果：若出现2个及以上1，则表示多数赞同，输出1

$\therefore F = AB + BC + AC = (A+B)(B+C)(C+A)$

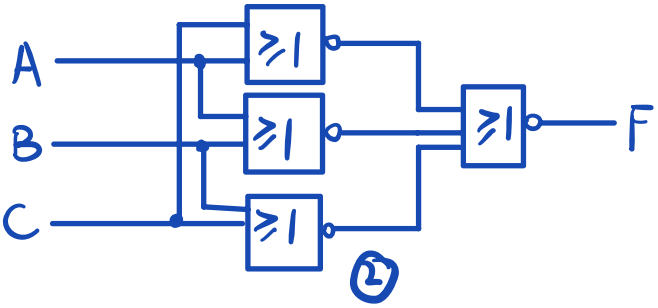
$= \overline{\overline{AB}\ \overline{BC}\ \overline{AC}}$ ①

$= \overline{\overline{(A+B)} + \overline{(B+C)} + \overline{(A+C)}}$ ②

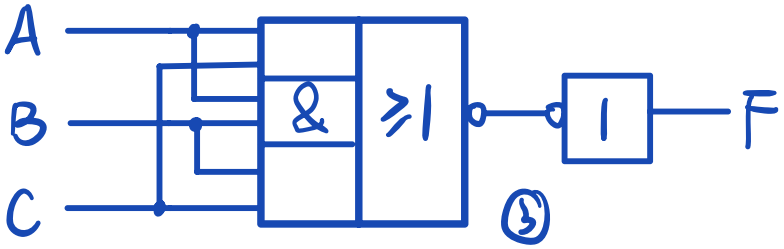
$= \overline{\overline{AB + BC + AC}}$ ③　$\overline{\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}}$



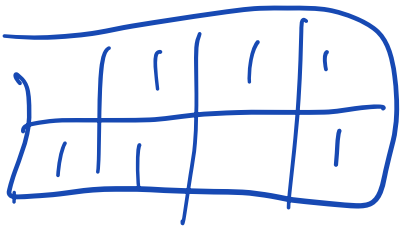
①



②



③



(2) 3 变量的非一致电路，以判断有不同意见。

输入：A，B，C　　输出：F

效果：若全为 0 或全为 1，则输出 0，否则输出 1

$$\therefore F = A\bar{B} + B\bar{C} + C\bar{A}$$

$$= \overline{A\bar{B}} \cdot \overline{B\bar{C}} \cdot \overline{C\bar{A}} = \bar{A}B \cdot \bar{B}C \cdot \bar{C}A$$

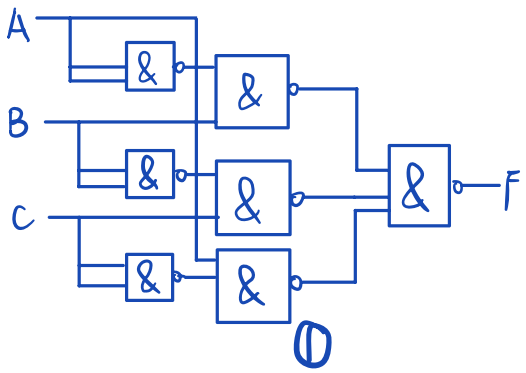
$$= \overline{(A+B+C)} + \overline{(\bar{A}+\bar{B}+\bar{C})}$$

$$= \overline{ABC} + \overline{\bar{A}\bar{B}\bar{C}}$$

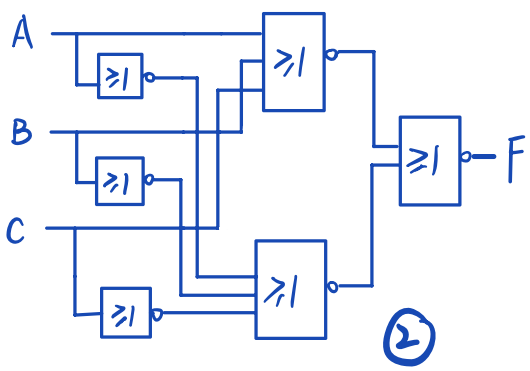
① $\overline{\overline{A\bar{B}}\ \overline{B\bar{C}}\ \overline{C\bar{A}}}$

② $\overline{\overline{\bar{A}+B} + \overline{\bar{B}+C} + \overline{A+\bar{C}}}$

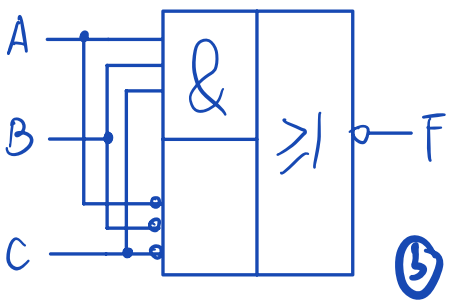
③ $\overline{A\bar{B} + \bar{A}C + B\bar{C}}$



①



②

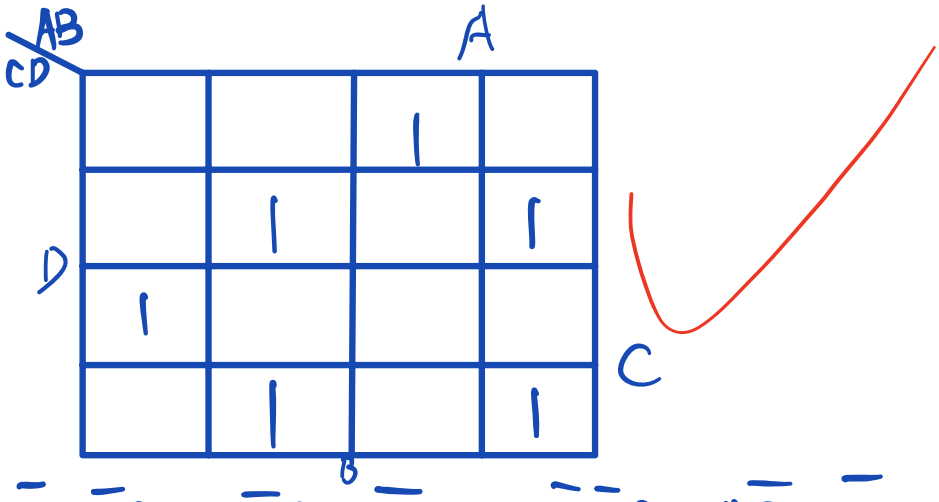


③

2.5(2) 分析：输入二进制数ABCD，输出F，若有两个1输出1，否则输出0

输出1的情况：1100、1010、1001、0110、0101、0011

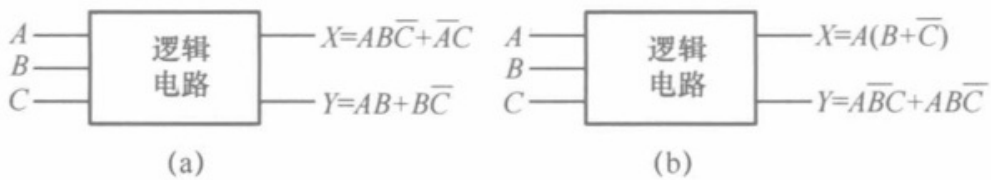
画卡诺图：



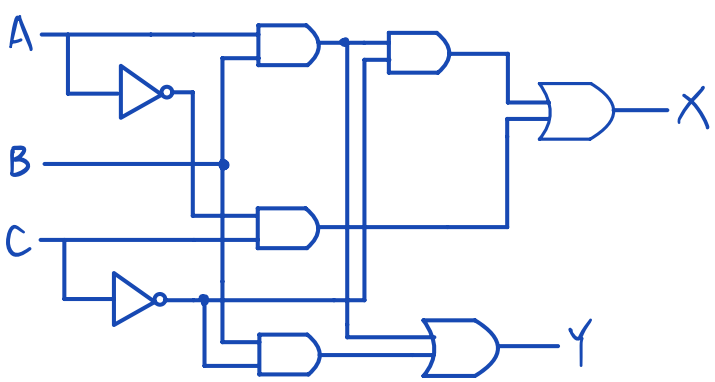
$F=\bar{A}\bar{B}CD+\bar{A}B\bar{C}D+\bar{A}BC\bar{D}+A\bar{B}\bar{C}D+A\bar{B}C\bar{D}+AB\bar{C}\bar{D}$

$=\sum m^4(3,5,6,9,10,12)$

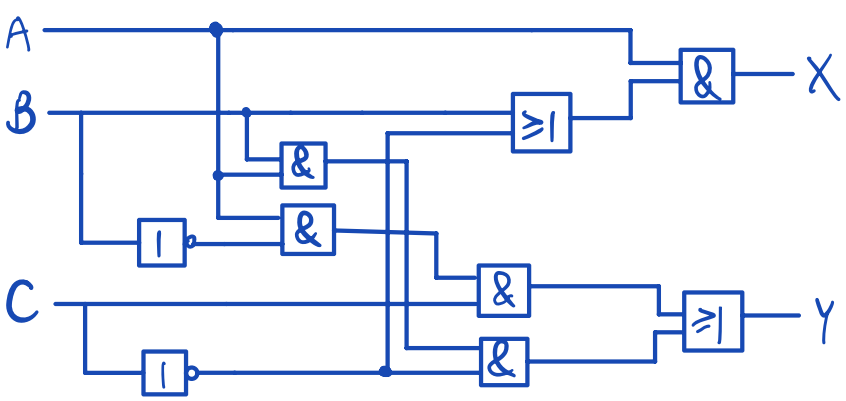
2.7



(a)



(b)



2.9 设计一个将典型 Gray 码转换为二进制编码的电路。

分析：输入 ABCD，输出 WXYZ

真值表：

A B C D	W X Y Z
0 0 0 0	0 0 0 0
0 0 0 1	0 0 0 1
0 0 1 1	0 0 1 0
0 0 1 0	0 0 1 1
0 1 1 0	0 1 0 0
0 1 1 1	0 1 0 1
0 1 0 1	0 1 1 0
0 1 0 0	0 1 1 1
1 1 0 0	1 0 0 0
1 1 0 1	1 0 0 1
1 1 1 1	1 0 1 0
1 1 1 0	1 0 1 1
1 0 1 0	1 1 0 0
1 0 1 1	1 1 0 1
1 0 0 1	1 1 1 0
1 0 0 0	1 1 1 1

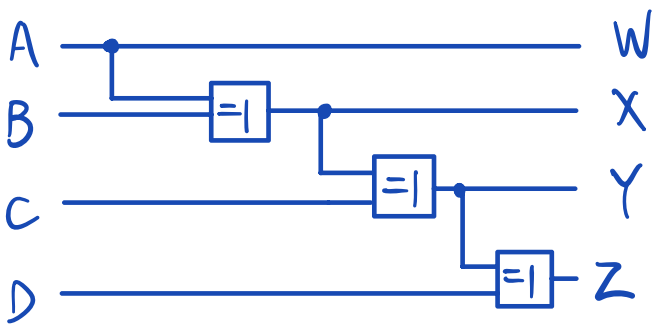
$\therefore W = A$

$X = A \oplus B$

$Y = A \oplus B \oplus C$

$Z = A \oplus B \oplus C \oplus D$

电路图：



2.11 用代数法判断下列函数是否存在逻辑险象，如果有的话，设法消除之。

(1) $F = \overline{A}B + \overline{B}\overline{C} + AC$

(2) $F = (A + C + \overline{D})(\overline{B} + C + D)(\overline{B} + \overline{C})(B + D)$

(1) ①当 $BC = 11$ 时，$F = \overline{A} + A$，存在静态 1 险象

②当 $AC = 00$ 时，$F = B + \overline{B}$，存在静态 1 险象

③当 $AB = 10$ 时，$F = \overline{C} + C$，存在静态 1 险象

消除：$F = \overline{A}B + \overline{B}\overline{C} + AC + \overline{A}\overline{C} + BC + A\overline{B}$

(2) ①当 $AD = 00$（$D=0$）或 010 或 100 或 110 时，$F = B \cdot \overline{B}$，存在静态 0 险象

②当 $AB = 01$ 时，$F = C \cdot \overline{C}$，存在静态 0 险象

③当 $AC = 00$ 或 010 时，$F = D \cdot \overline{D}$，存在静态 0 险象

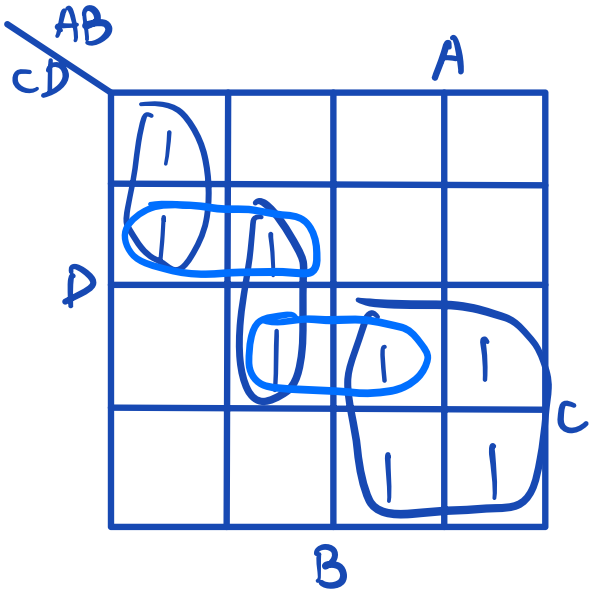
消除：$F = (A + C + \overline{D})(\overline{B} + C + D)(\overline{B} + \overline{C})(B + D)(\overline{A} + C + D)(\overline{B} + D)(C + D)(\overline{C} + D)(A + C)(A + \overline{B} + \overline{D})(A + \overline{B} + C)$

2.12 用卡诺图化简下列函数，所得函数中不得有逻辑险象。

（1）$F=\sum m^4(0,1,5,7,10,11,14,15)$

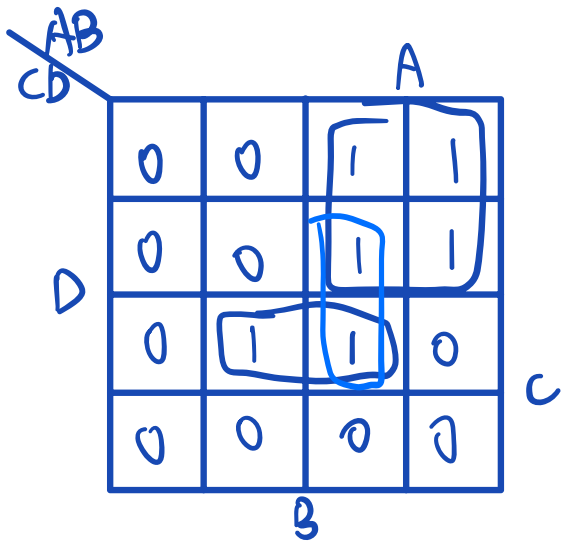
（2）$F=\prod M^4(0,1,2,3,4,5,6,10,11,14)$

(1)



$F=\bar{A}\bar{B}\bar{C}+\bar{A}BD+AC+\bar{A}\bar{C}D+BCD$

(2)



$F=A\bar{C}+BCD+\bar{A}BD$